Week 8

Question 1 Regression is an example of –
A. Supervised learning
B. Unsupervised learning
C. Clustering
D. Classification

A. Identify the Core Concept Being Tested The fundamental classification of machine learning tasks, specifically categorizing regression among supervised and unsupervised learning frameworks.

B. Rigorous Solution The correct answer is **A. Supervised learning**.

C. Derivations / Mathematical Reasoning No mathematical derivation is required for categorizing machine learning tasks. In data mining, tasks are broadly divided based on the presence of labeled training data. Regression algorithms strictly require a training set where every example consists of input attributes paired with a known, predefined target value. Because this explicit target acts as "supervision" guiding the learning process, regression is inherently a supervised learning task.

E. Concept Expansion Supervised learning tasks rely on training data that contains explicit "class level information" or target outputs. The goal is predictive mining: using the patterns learned from the supervised training set to predict the outcome for an unseen future example. This directly contrasts with unsupervised learning (like clustering), where the algorithm receives attribute values without any corresponding predefined targets and must independently discover homogeneous groups in an exploratory manner.

Question 2 The output of a regression function is usually –
A. Categorical valued
B. Real valued
C. Negative valued
D. Positive valued

A. Identify the Core Concept Being Tested The defining characteristic that distinguishes regression tasks from classification tasks based on the mathematical nature of the target output.

B. Rigorous Solution The correct answer is **B. Real valued**.

C. Derivations / Mathematical Reasoning The defining mathematical property of regression is its domain of predictions. A regression model $f(x;\theta)$ specifically targets the prediction of a continuous variable $y \in \mathbb{R}$. The course explicitly states that in regression, the

"output is not a class – output is a real value". Examples include predicting rainfall in cm or predicting future stock prices.

E. Concept Expansion The type of dependent variable dictating the task is a core principle in data mining. If the dependent variable maps to discrete, predefined categories (e.g., spam vs. non-spam, sports vs. politics), the problem is formally Classification. If the dependent variable spans an infinite, continuous numerical spectrum (real-valued), the problem is formally mapped as Regression.

Question 3 If x_1, x_2 are independent variables and y is the dependent variable, which of the following represents a linear regression model?

- **A.** $y = a_0 + \frac{a_1}{x_1} + \frac{a_2}{x_2}$
- **B.** $y = a_0 + a_1x_1 + a_2x_2$
- **C.** $y = a_0 + a_1x_1 + a_2x_2^2$
- **D.** $y = a_0 + a_1x_1^2 + a_2x_2$

A. Identify the Core Concept Being Tested The mathematical formulation and standard notation of a multivariate linear regression model.

B. Rigorous Solution The correct answer is **B.** $y = a_0 + a_1x_1 + a_2x_2$.

C. Derivations / Mathematical Reasoning

1. Let the dependent output variable be y and the independent input variables be mapped to a vector $X = [x_1, x_2, \ldots, x_k]$.
2. A regression model attempts to map $y = f(x_1, x_2, \ldots, x_k)$.
3. For the model to strictly qualify as "linear", the function f must consist solely of a linear combination of the input variables weighted by regression coefficients (parameters).
4. The general analytical form for a multivariate linear regression model is given as $y = a_0 + a_1x_1 + a_2x_2 + \cdots + a_kx_k$.
5. For a two-variable system ($k = 2$), this equation truncates exactly to $y = a_0 + a_1x_1 + a_2x_2$. Options featuring $1/x_1$ or x^2 inherently introduce non-linear relationships and thus violate the constraint of a purely linear model.

E. Concept Expansion In a 2-dimensional feature space, the mathematical equation $y = a_0 + a_1x_1 + a_2x_2$ geometrically represents a flat 2D plane existing within a 3D space. The parameter a_0 acts as the bias or axis intercept, while a_1 and a_2 dictate the respective slope of the regression plane along the x_1 and x_2 axes. Because it restricts the hypothesis space to linear weights, it is easy to interpret and design, often serving as a powerful baseline model.

Question 4 The output of a regression model is the –

A. Independent variable

B. Dependent variable

C. Integer variable

D. Null variable

A. Identify the Core Concept Being Tested The standard statistical terminology mapping inputs and outputs within a functional regression equation.

B. Rigorous Solution The correct answer is **B. Dependent variable**.

C. Derivations / Mathematical Reasoning By standard mathematical definition in the regression function $y = f(x_1, x_2, \ldots, x_k)$, the predicted output y intrinsically relies on the combination of the inputs. Consequently, the regression output y is universally termed the "dependent variable". The input attributes $x_1, x_2, \ldots, x_k$ are the quantities that are manipulated or observed independently, rendering them the "independent variables".

E. Concept Expansion The entire premise of the model f is to mathematically determine exactly how the dependent variable y functionally depends upon the changes within the independent variables x. This terminology highlights the directional flow of cause-and-effect estimation central to predictive modeling.

Question 5 Which of the following is most commonly used as an error measure in linear regression?

A. Mean squared error

B. Classification error

C. Scatter coefficient

D. Silhouette coefficient

A. Identify the Core Concept Being Tested The mathematical "Score function" optimized during the training phase of a standard linear regression model.

B. Rigorous Solution The correct answer is **A. Mean squared error**.

C. Derivations / Mathematical Reasoning

1. During training, a model evaluates its fitness by calculating the deviation between the actual target values $y(i)$ and the predicted values $f(x(i); \theta)$.
2. The fundamental error term for a single instance is $e_i = y_i - f(x_i; \theta)$.
3. Because deviations can be both positive (overshooting) and negative (undershooting), simply summing e_i causes cancellations. To ensure all errors penalize the model equally, the error magnitude is squared: e_i^2.
4. Aggregating these across the dataset yields the Sum Squared Error (SSE): $S(\theta) = \sum e_i^2$. Dividing by the number of data points N yields the Mean Squared Error (MSE), making it

the ubiquitous score function optimized via algorithms like gradient descent or direct matrix inversion.

E. Concept Expansion In data mining architecture, selecting the "Score function" is a core component. Minimizing squared errors yields the unique mathematical advantage of producing a smooth, differentiable convex quadratic cost function. This guarantees that taking the analytical derivative of the score function with respect to the parameters yields linear Normal Equations ($X'X\theta = X'y$), allowing for reliable global minimum optimization using standard linear algebra.

Question 6 The number of input variables in linear regression can be -
A. One
B. Two
C. Three
D. Many

A. Identify the Core Concept Being Tested The flexible scalability of the input dimensionality in regression architecture.

B. Rigorous Solution The correct answer is **D. Many**.

C. Derivations / Mathematical Reasoning While a regression model featuring exactly one input variable is specifically termed "Univariate regression" ($y = a_0 + a_1 x$), standard regression frameworks natively expand to "Multivariate regression" where the prediction model accepts multiple input variables ($y = f(x_1, x_2, \ldots, x_k)$). Therefore, the quantity of input independent variables is not constrained to one, two, or three, but can easily be "Many".

E. Concept Expansion While theoretically, there can be an infinitely large p number of variables, utilizing "many" variables introduces mathematical vulnerabilities in practice. If the number of variables p grows extremely large, the computational complexity $O(Np^2 + p^3)$ for direct matrix inversion skyrockets. Furthermore, models with too many independent parameters face the "Curse of Dimensionality" and severe collinearity issues, which necessitates upstream dimensionality reduction steps like Feature Selection or PCA.

Question 7

The linear regression model $y = a_0 + a_1x$ is applied to the data in the table shown below. What is the value of the mean squared error, when $a_0 = 0$, $a_1 = 1$?

x	y
0	1
1	2
2	3
2.5	3.5

A. 0.00

B. 0.25

C. 0.50

D. 1.00

A. Identify the Core Concept Being Tested Numerical computation of the Mean Squared Error (MSE) over a discrete dataset using a specified univariate linear regression model.

B. Rigorous Solution The correct answer is **D. 1.00**.

D. Numerical Solution

- **Governing Equation:** For a dataset of size N, the Mean Squared Error is $MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$, where predicted value $\hat{y}_i = a_0 + a_1x_i$.
- **Substitute Values:** The model is fixed at $a_0 = 0$ and $a_1 = 1$, making our prediction equation strictly $\hat{y}_i = 1 \cdot x_i = x_i$. The dataset contains $N = 4$ points.
- **Compute intermediate results (point-by-point):**
  1. Point 1: $x = 0$, Actual $y = 1$. Pred $\hat{y} = 0$. Error $e_1 = 1 - 0 = 1$. Squared Error $e_1^2 = 1^2 = 1$.
  2. Point 2: $x = 1$, Actual $y = 2$. Pred $\hat{y} = 1$. Error $e_2 = 2 - 1 = 1$. Squared Error $e_2^2 = 1^2 = 1$.
  3. Point 3: $x = 2$, Actual $y = 3$. Pred $\hat{y} = 2$. Error $e_3 = 3 - 2 = 1$. Squared Error $e_3^2 = 1^2 = 1$.
  4. Point 4: $x = 2.5$, Actual $y = 3.5$. Pred $\hat{y} = 2.5$. Error $e_4 = 3.5 - 2.5 = 1$. Squared Error $e_4^2 = 1^2 = 1$.
- Sum of Squared Errors (SSE) $= 1 + 1 + 1 + 1 = 4$.
- Mean Squared Error (MSE) $= \frac{SSE}{N} = \frac{4}{4} = 1.00$.
- **Final Value Interpretation:** Given these parameters, the model chronically underpredicts the target by exactly 1 unit everywhere, resulting in a uniform squared error of 1.00 per data instance.

E. Concept Expansion This exercise highlights how unoptimized parameters yield a high systematic error (bias). Because the error magnitude across the dataset is uniform ($e_i = 1$ everywhere), the model's chosen slope ($a_1 = 1$) perfectly captures the variance of the data,

but the bias or intercept (a_0) is mathematically shifted. Optimization algorithms (like gradient descent) would instantly utilize this steady error gradient to adjust the a_0 parameter upward to close the gap.

Question 8

The linear regression model $y = a_0 + a_1x$ is to be fitted to the data in the table shown below. What is the optimal regression model obtained by minimizing sum squared error?

x	y
0	1
1	2
2	3
2.5	3.5

A. $y = 1.01 - 2.10x$

B. $y = 1.00 + 2.10x$

C. $y = 1.01 - 1.00x$

D. $y = 1.00 + 1.00x$

A. Identify the Core Concept Being Tested Deducing the optimal set of linear coefficients (a_0, a_1) that functionally drops the squared error to its theoretical global minimum.

B. Rigorous Solution The correct answer is **D.** $y = 1.00 + 1.00x$.

D. Numerical Solution

- **Governing Concept:** The optimal regression model minimizes $S(\theta) = \sum[y_i - (a_0 + a_1x_i)]^2$.
- **Analyze Dataset:** $x = 0 \implies y = 1$ $x = 1 \implies y = 2$ $x = 2 \implies y = 3$ $x = 2.5 \implies y = 3.5$
- **Compute intermediate results:** Subtract x from y for each instance: $1 - 0 = 1$ $2 - 1 = 1$ $3 - 2 = 1$ $3.5 - 2.5 = 1$
- The relationship is demonstrably collinear and perfect; for every point, y is exactly equal to $x + 1$.
- **Substitute optimal values:** Mapping this physical reality to the univariate format $y = a_0 + a_1x$, it structurally necessitates that the slope intercept $a_0 = 1.00$ and the slope $a_1 = 1.00$.
- **Final Value Interpretation:** If we evaluate $S(\theta)$ using equation $y = 1.00 + 1.00x$, the predicted $\hat{y}_i$ values will equal the target y_i values perfectly at every coordinate. The Sum Squared Error falls absolutely to 0.00, confirming it is the strict mathematical optimum.

E. Concept Expansion While this problem yields a trivial empirical inspection due to perfect correlation, standard data pipelines compute this analytically by solving the Normal

Equations $\theta = (X'X)^{-1}X'y$. When independent and dependent variables are perfectly correlated, the mapping acts as an exact linear interpolation, shifting the optimal empirical training set error (and bias) entirely to zero.

Question 9

The linear regression model $y = a_0 + a_1x_1 + a_2x_2$ is applied to the data in the table shown below. What is the value of the mean squared error, when $a_0 = 0$, $a_1 = 1$, $a_2 = 1$?

x_1	x_2	y
0	0	1
0	1	2
1	0	2
1	1	3

A. 0.00

B. 0.25

C. 0.50

D. 1.00

A. Identify the Core Concept Being Tested Numerical evaluation of the Mean Squared Error for a Multivariate (two independent variable) Linear Regression framework.

B. Rigorous Solution The correct answer is **D. 1.00**.

D. Numerical Solution

- **Governing Equation:** $MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$, where the multivariate prediction is given by $\hat{y}_i = a_0 + a_1x_{1i} + a_2x_{2i}$.
- **Substitute Values:** The given parameters are $a_0 = 0$, $a_1 = 1$, and $a_2 = 1$. The functional equation becomes $\hat{y} = 0 + 1(x_1) + 1(x_2) = x_1 + x_2$.
- **Compute intermediate results (point-by-point):**
  1. Point $(0, 0)$: Actual $y = 1$. Pred $\hat{y} = 0 + 0 = 0$. Error $e_1 = 1 - 0 = 1$. Sq Error $e_1^2 = 1$.
  2. Point $(0, 1)$: Actual $y = 2$. Pred $\hat{y} = 0 + 1 = 1$. Error $e_2 = 2 - 1 = 1$. Sq Error $e_2^2 = 1$.
  3. Point $(1, 0)$: Actual $y = 2$. Pred $\hat{y} = 1 + 0 = 1$. Error $e_3 = 2 - 1 = 1$. Sq Error $e_3^2 = 1$.
  4. Point $(1, 1)$: Actual $y = 3$. Pred $\hat{y} = 1 + 1 = 2$. Error $e_4 = 3 - 2 = 1$. Sq Error $e_4^2 = 1$.
- Sum of Squared Errors (SSE) $= 1 + 1 + 1 + 1 = 4$.
- There are $N = 4$ instances. Mean Squared Error (MSE) $= \frac{SSE}{N} = \frac{4}{4} = 1.00$.
- **Final Value Interpretation:** Even as the model scales across an additional dimension (x_2), the uniform bias error logic persists. A parallel plane offset vertically by exactly 1 unit yields a uniform point-plane spatial error of 1.

E. Concept Expansion This process maps exactly to finding the vertical spatial distances spanning between scattered 3-dimensional data points and a 2-dimensional regression plane. A non-zero error validates that the plane $y = x_1 + x_2$ is not the optimal geometrical sub-space fitting the data structure; mathematically altering the bias intercept a_0 to 1 would snap the plane precisely into the coordinates, forcing the structural error to 0.

Question 10 We consider Regression Model A to be a linear regression line, and Regression Model B to be a cubic regression curve. Which of the following statement is true when these models are trained on a small data set?

A. Model A might underfit, and Model B might overfit
B. Model A might overfit, and Model B might underfit
C. Model A is more complex than Model B
D. Model A will have higher training set error than Model B

A. Identify the Core Concept Being Tested The fundamental relationship balancing model complexity against generalization capability, encapsulating the phenomenon of "overfitting".

B. Rigorous Solution The correct answer is **A. Model A might underfit, and Model B might overfit**.

C. Derivations / Mathematical Reasoning

1. Model complexity represents the degrees of freedom mathematically available to a model. Model A (linear line) has just two degrees of freedom (a_0, a_1). Model B (cubic curve) possesses four degrees of freedom ($a_0, a_1x, a_2x^2, a_3x^3$).
2. Occam's razor posits that applying an overly complex model (cubic curve) to a minor, small dataset allows the curve to twist wildly to match every random noise artifact mapping the training set flawlessly. This rote memorization without actual trend discovery is structurally defined as *overfitting* (characterized mathematically by low bias but massively high variance).
3. Conversely, if the underlying phenomenon generating the data contains native non-linearity, a simple linear model (Model A) lacks the degrees of freedom to ever map the true shape. It assumes too generic of a representation and will consequently *underfit* the relationship (displaying high structural bias error).

E. Concept Expansion This question isolates the central tradeoff of all predictive data mining. We are not interested strictly in minimizing the error purely on the training data (empirical learning); we desire the model to securely predict unseen future data (generalization). Driving training set error identically to zero utilizing overly complex models invariably catastrophically corrupts test set error. The overarching goal is stopping at the optimal inflection point on the complexity curve right before test error surges.

Question 11 Autoregressive models for a time series predict –

A. Values of future time steps from past time step values

B. Values of past time steps from future time step value

C. Values of present time step present time step value

D. Values of present time step from a different regression model

A. Identify the Core Concept Being Tested The structural inputs and targets governing Autoregressive models executed for time-series predictions.

B. Rigorous Solution The correct answer is **A. Values of future time steps from past time step values**.

C. Derivations / Mathematical Reasoning

1. A conventional time-series consists of temporal measurements tracked sequentially: $x_1, x_2, \ldots, x_t$.
2. The predictive regression task involves deducing the unknown future state x_{t+1} by projecting from the known history.
3. In an Autoregressive Model, the regression function treats the sequential historical values natively as the "independent variables" and the future unknown value as the "dependent variable":

$$x_{t+1} = f(x_1, \ldots, x_t; \theta) = \sum_k \alpha_k x_{t-k}$$

.

4. Because the model strictly regresses a variable against delayed chronological versions of *itself*, it uniquely predicts the "values of future time steps from past time step values".

E. Concept Expansion Autoregressive techniques creatively retrofit classical static regression architectures to capture dynamic temporal progression. By treating history sequences conceptually exactly like an upstream multivariate feature matrix (X), time-series analysis instantly reaps the robust mathematical advantages of squared-error optimization, allowing direct matrix algebra execution to solve linear coefficients mapping chronological memories.